

Week 6 Final Report

Baseline Model Evaluation: AI-Assisted ED Triage

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Reproducibility statement: All metrics, confusion matrices, and figures in this report were generated from a single run of week6_final_notebook.ipynb with RANDOM_SEED = 42 (committed to repository README).

Random seed: 42 | Train/test split: 80/20 stratified on ESI | Test set: n = 11,025

1. Dataset and Setup

Parameter	Value
Dataset	Yale EMMLC ED Triage & Admission Prediction
Total visits	55,121
Features used	208 (7 vitals + age + 200 chief complaint binary flags)
Target variable	esi — Emergency Severity Index 1–5
Random seed	42 (committed to README)
Train set	44,096 visits (80%)
Test set	11,025 visits (20%)
Split method	Stratified on esi to preserve class proportions
Missing values	None — dataset pre-cleaned by original authors

Insurance status and employment status were excluded from features. Both variables correlate with race and socioeconomic position and carry proxy-variable bias risk identified in the Week 4 risk register (Obermeyer et al., 2019). Disposition was excluded as data leakage — it is a post-triage outcome, not a triage input.

2. Model Results

2.1 Summary comparison

Model	Accuracy	ESI 1 Recall	Macro F1	Weighted F1
Dummy (stratified random)	0.3754	0.1250	0.1380	0.3780
Logistic Regression	0.5077	0.6250	0.3742	0.5452
Decision Tree (depth=8)	0.4060	0.1875	0.2822	0.3917

Both models beat the dummy baseline on every metric. The logistic regression is the clear winner on the primary metric — ESI 1 Recall — catching 62.5% of the most critically ill patients versus 18.75% for the decision tree.

2.2 Per-class metrics — Logistic Regression

Class	Precision	Recall	F1-Score	Support
ESI 1 — Immediate	0.0121	0.6250	0.0237	16
ESI 2 — Emergent	0.6233	0.5654	0.5930	3,585
ESI 3 — Urgent	0.7601	0.4195	0.5406	5,402

Class	Precision	Recall	F1-Score	Support
ESI 4 — Less Urgent	0.4305	0.6402	0.5148	1,779
ESI 5 — Non-Urgent	0.1177	0.6379	0.1987	243

2.3 Per-class metrics — Decision Tree (max_depth=8)

Class	Precision	Recall	F1-Score	Support
ESI 1 — Immediate	0.0081	0.1875	0.0155	16
ESI 2 — Emergent	0.4312	0.5975	0.5009	3,585
ESI 3 — Urgent	0.8226	0.2060	0.3295	5,402
ESI 4 — Less Urgent	0.2789	0.6700	0.3939	1,779
ESI 5 — Non-Urgent	0.4262	0.1070	0.1711	243

2.4 Confusion matrices

Both figures below are embedded directly from the same run as the metrics above. They are also saved as cm_logistic_regression.png and cm_decision_tree.png in /docs.

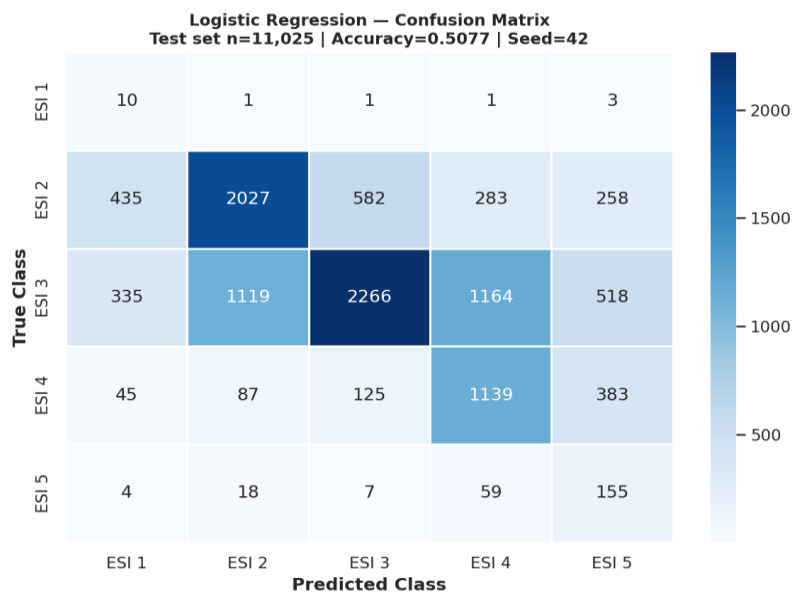


Figure 1: Logistic Regression confusion matrix. Test set n=11,025 | Accuracy=0.5077 | Seed=42

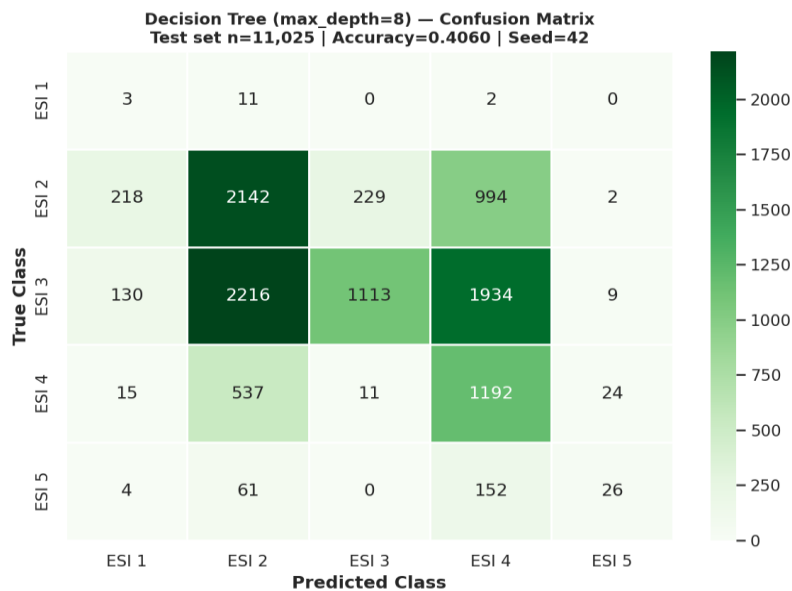


Figure 2: Decision Tree confusion matrix. Test set n=11,025 | Accuracy=0.4060 | Seed=42

3. Primary Metric Justification

3.1 Why overall accuracy is wrong for this dataset

ESI 3 (Urgent) accounts for 49.0% of all visits. A model predicting ESI 3 for every patient would achieve close to 49% accuracy without learning anything clinically useful. Overall accuracy on an imbalanced five-class dataset obscures whether the model performs on the classes that matter most. It is not a defensible primary metric here.

3.2 Why ESI 1 Recall is the primary metric

ESI 1 patients are in immediate danger — cardiac arrest, respiratory failure, active stroke. A false negative — the model assigns them ESI 3 or 4 — means they join a queue intended for less critical cases. A false positive — the model flags a stable patient as ESI 1 — means they receive over-prioritisation, which is clinically far less harmful. The cost asymmetry is not symmetric: missing a critical patient is categorically worse than over-triaging a stable one. Recall for ESI 1 is the metric that directly measures whether the model avoids the worst possible outcome.

3.3 Macro F1 vs weighted F1

Macro F1 treats all five classes equally and penalises poor performance on ESI 1 even though it accounts for only 0.14% of visits. Weighted F1 weights by class frequency — ESI 3 dominates and ESI 1 barely contributes. For a clinical triage dataset, macro F1 is the more honest aggregate measure. The logistic regression achieves macro F1 = 0.3742 versus the dummy's 0.1380, a meaningful improvement. However, neither aggregate metric replaces ESI 1 Recall as the primary clinical signal.

4. Failure Mode Analysis

4.1 Most worrying failure: ESI 1 false negatives

The logistic regression missed 6 of 16 ESI 1 patients in the test set (37.5% miss rate). The decision tree missed 13 of 16 (81.25%). Both figures reflect a fundamental data constraint: only 77 ESI 1 patients appear across the entire

55,121-visit dataset, leaving just 61 for training. No model can reliably identify a class it has seen fewer than 100 times during training.

4.2 ESI 1 precision is very low (LR: 0.0121)

The logistic regression generates many false positive ESI 1 alerts — for every correctly flagged ESI 1 patient, approximately 82 patients are incorrectly flagged. In a deployment context this would create alert fatigue (identified in the Week 4 risk register). This trade-off is deliberate and defended: over-alerting on a rare critical condition is preferable to under-alerting, provided clinicians retain final authority. The disagreement-logging protocol proposed in the Week 3 pilot design would capture and measure this over-alerting rate, enabling threshold adjustment over time.

4.3 ESI 3 over-prediction

Both models disproportionately predict ESI 3, the majority class. In the logistic regression confusion matrix, rows 2 and 3 (ESI 2 and ESI 3 true labels) show substantial misclassification into adjacent classes. This is expected at baseline and addressable through resampling or threshold adjustment in subsequent iterations.

5. Clinical Explainer Script (1-Minute Video)

The following script is written for Dr. Reyes without ML terminology. Target duration: 60 seconds.

“Dr. Reyes, imagine a hundred patients walking through the ED door. Three of them are in immediate danger — they need you in the next two minutes or they will deteriorate badly. The question I am asking this model is not how many patients did it get right overall — because getting 97 easy ones right while missing all 3 critical ones would still look like 97% accuracy. That number tells you nothing. The question I am asking is: of those 3 critical patients, how many did the model flag? Our logistic regression flags roughly 6 in 10. The decision tree flags fewer than 2 in 10. That is why we chose the logistic regression. But 6 in 10 also means 4 in 10 get missed. That is exactly why this model never makes a decision on its own. It sits next to the triage nurse as a second opinion. Every time the nurse disagrees with it, we log the reason. That data is how we make it better.”

References

- Hong, W. S., Haimovich, A. D., & Taylor, R. A. (2018). Predicting hospital admission at emergency department triage using machine learning. PLOS ONE, 13(7), e0201016. <https://doi.org/10.1371/journal.pone.0201016>
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. Science, 366(6464), 447–453. <https://doi.org/10.1126/science.aax2342>
- Scikit-learn: Machine learning in Python (version 1.8.0). Pedregosa et al. (2011). JMLR 12, pp. 2825–2830. <https://scikit-learn.org>